Duppalapudi Kamal Koushik

ch23btech11016@iith.ac.in | kamalkoushik.d@gmail.com | +91 9963252792 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.32
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2023	94.70%
Sri Chaitanya High School	Class X (State Board of Secondary Education, Andhra Pradesh)	2021	100.00%

SCHOLASTIC ACHIEVEMENTS

369 / 11,517, PicoCTF 2024 - Global cybersecurity CTF by Carnegie Mellon University.

PROFESSIONAL EXPERIENCE

- **Lead of Software Engineering & Product Architect, semXai Technologies**

- Run all company compute: an **8×A100 and L40S GPU fleet** plus the full **Google Cloud** estate, owning provisioning, capacity planning, access control, and spend for every engineering team.

- Designed **multi-GPU orchestration** for the TensorTune benchmarking platform with singleton **vLLM** engine slots and **GPU bin-packing**, letting concurrent jobs share the fleet without OOM failures.

- Architect **WattWise**, an AI-driven DVFS system for GPUs and CPUs: a **Linux kernel module** and **C daemon** running **ONNX** models in-process over a **ZMQ/Protobuf** telemetry pipeline, saving up to **30%** energy on CPU workloads.

- Serve as cybersecurity advisor, setting security review and access practice across all products and environments.

- **L1 SOC Analyst, SEQRITE Labs, Quick Heal Technologies Limited**

- Triaged and analysed incidents across **EPP, EDR/XDR, email, and firewall telemetry** on the SEQRITE XDR platform, mapping adversary TTPs to **MITRE ATT&CK**.

- Evaluated and improved **SIA**, the platform's Gen AI agentic module for autonomous incident closure, through model evaluation, multi-step action chaining, and edge-case identification.

- Raised autonomous closure accuracy and tightened escalation boundaries, cutting repetitive Tier-1 analyst workload.

- **DevOps Intern, ITC Infotech Limited**

- Integrated **Azure Databricks Asset Bundles** into existing production pipelines to streamline ML workflow deployment and versioning.

- Automated continuous integration and deployment using **Azure DevOps Pipelines** for scalable data processing jobs.

- Worked with **Azure ML Studio** to manage model lifecycle and integrated it with Databricks for end-to-end MLOps.

- Collaborated with data engineering teams to ensure seamless data ingestion and transformation across services.

POSITIONS OF RESPONSIBILITY

Deputy Contingent Leader, Inter IIT Tech Meet 14.0, IIT Patna

- Led the IIT Hyderabad contingent of **14 teams (96 members)** across AI/ML, robotics, networking, and product design.

- Ran team selection, mentor allocation, and submission review cycles, and owned preparation timelines and logistics.

Overall Head, Kludge, InfoSec & Networking Club, IITH

- Led an **18-member** core team and oversaw **5+** cybersecurity projects; ran **MilanCTF24** and **EnigmaCTF25**.

- Designed CTF challenges and hosting infrastructure, and delivered workshops on networking, secure coding, and vulnerability analysis.

SKILLS

Cybersecurity: Netcat, Nmap, Burp Suite, Hashcat, Aircrack-ng, Wireshark, Socat, Bettercap

Infrastructure: Linux (Fedora, Arch), Git, NVIDIA Triton, vLLM, GStreamer, gRPC, ZMQ, Protobuf, CMake, LAN design, DNS

Programming: Python, C/C++, Go, Bash, SQL, TypeScript

DevOps/Cloud: Docker, GitHub Actions, Azure DevOps, Azure (Databricks, ML Studio), Google Cloud, AWS EC2, Oracle OCI, Cloudflare, Nginx

PROJECTS

- **Real-Time Video Surveillance & NVR Platform**

- Built a platform ingesting RTSP feeds from **200+ PTZ, IP, and ANPR cameras**, using camera substreams for analytics and main streams for recording, keeping decode cost flat as the fleet grows.

- Ran **GStreamer** hardware-accelerated decode with **NVIDIA Triton Inference Server** and per-model batching on **A100/H100** GPUs, with an SFU layer fanning one decoded stream to many clients.

- Built a **Go + gRPC** control plane with audit logging and encryption at rest to meet **DPDP Act** requirements.

- **Fuse: Custom Transport Protocol (C++, UDP)**

- Built a general-purpose transport over raw datagram sockets with per-stream **LOSSLESS / ORDERED / COALESCE** flags, so applications choose their own reliability and ordering tradeoffs.

- Implemented fixed-size circular stream windows with **bitmask ACK/NACK** loss detection, a three-way **SETUP handshake** pinning streams to worker threads, and optional **DTLS (wolfSSL)**.

RELEVANT COURSES

Computer Science: Intro to Programming, Intro to AI, Harvard CS50X/CS50P/CS50T, Data Structures and Algorithms

Systems: Computer Networks, Computer Architecture

EXTRACURRICULAR

1st place, MilanCTF 2023 - IIT Hyderabad's Inter-Hostel Cybersecurity Championship.

1st place, EnigmaCTF 2024 - Techno-Cultural Fest CTF, IIT Hyderabad.